

Graph neural networks for automatic extraction and labeling of the coronary artery tree in CT angiography

Nils Hampe^{a,b,c,*} Sanne G. M. van Velzen^{a,b,c} Jelmer M. Wolterink^{a,d}
Carlos Collet^e José P. S. Henriques^f Nils Planken^g and Ivana Išgum^{a,b,c,g}

^aAmsterdam University Medical Center location University of Amsterdam,

Department of Biomedical Engineering and Physics, Amsterdam, The Netherlands

^bAmsterdam Cardiovascular Sciences, Amsterdam, The Netherlands

^cUvA, Informatics Institute, Faculty of Science, Amsterdam, The Netherlands

^dUniversity of Twente, Technical Medical Centre, Department of Applied Mathematics, Enschede, The Netherlands

^eOLV Clinic, Cardiovascular Center, Aalst, Belgium

^fAmsterdam University Medical Center location University of Amsterdam, AMC Heart Center, Amsterdam, The Netherlands

^gAmsterdam University Medical Center location University of Amsterdam, Department of Radiology and Nuclear Medicine, Amsterdam, The Netherlands

ABSTRACT. **Purpose:** Automatic comprehensive reporting of coronary artery disease (CAD) requires anatomical localization of the coronary artery pathologies. To address this, we propose a fully automatic method for extraction and anatomical labeling of the coronary artery tree using deep learning.

Approach: We include coronary CT angiography (CCTA) scans of 104 patients from two hospitals. Reference annotations of coronary artery tree centerlines and labels of coronary artery segments were assigned to 10 segment classes following the American Heart Association guidelines. Our automatic method first extracts the coronary artery tree from CCTA, automatically placing a large number of seed points and simultaneous tracking of vessel-like structures from these points. Thereafter, the extracted tree is refined to retain coronary arteries only, which are subsequently labeled with a multi-resolution ensemble of graph convolutional neural networks that combine geometrical and image intensity information from adjacent segments.

Results: The method is evaluated on its ability to extract the coronary tree and to label its segments, by comparing the automatically derived and the reference labels. A separate assessment of tree extraction yielded an *F1* score of 0.85. Evaluation of our combined method leads to an average *F1* score of 0.74.

Conclusions: The results demonstrate that our method enables fully automatic extraction and anatomical labeling of coronary artery trees from CCTA scans. Therefore, it has the potential to facilitate detailed automatic reporting of CAD.

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*Address all correspondence to Nils Hampe, n.hampe@amsterdamumc.nl

1 Introduction

Coronary artery disease (CAD) is the leading cause of death worldwide.¹ It is characterized by a buildup of atherosclerotic plaque in the coronary artery wall that may lead to obstruction in the blood supply to the heart muscle. Coronary CT angiography (CCTA) enables visualization of the coronary arteries and determination of the presence and type of atherosclerotic plaque and grade of stenosis. In this process, clinical reporting of CAD requires characterization of plaque burden and quantification of the stenosis grade, along with its anatomical location.² As detailed reporting performed by an expert is time-consuming and expensive, routine clinical reporting is currently restricted to the largest artery segments, determined by visual or semi-automatic assessment.³ Automatic analysis and reporting of CAD would relieve the expert of this workload. Moreover, detailed automatic reporting that would include all anatomical segments of the coronary artery tree may broaden research regarding the pathophysiology of segment-specific plaque formation,⁴ and clinical risk stratification.^{5,6}

To build a coronary artery tree, coronary artery centerlines are typically extracted first. Early methods for the extraction of the coronary artery centerlines found the minimum path between manually or automatically defined start and end points on a predefined cost map^{7,8} or applied thinning operations on initial vessel lumen segmentations.^{9,10} More recently, methods have been proposed that use a tracker that predicts local vessel directions to obtain centerlines from automatically or manually placed seed points.^{11–14} For this, Lesage et al.¹¹ used kernel density estimation to learn a geometrical model based on handcrafted features. To directly use image data as input to the tracker, Wolterink et al.¹² proposed to train a convolutional neural network (CNN) for predicting the tracking direction. Salahuddin et al.¹³ demonstrated that using multi-resolution input to the CNN-based tracker improves the performance of coronary artery centerline extraction.

Downstream methods that require information about relationships between arteries, e.g., to label its segments, typically construct a coronary artery tree. To build the coronary artery tree from separately extracted centerlines, information regarding the relationships among individual centerlines is required. Previously developed methods stored this information in a tree graph.^{15,16}

Alternatively, the tree graph can be built during simultaneous extraction of the entire coronary artery tree. For this, Salahuddin et al.¹³ initialized additional trackers for side branches at automatically detected bifurcations.

Building the coronary artery tree with these methods requires accurate extraction of the coronary artery centerlines. Previous methods for automatic coronary centerline extraction were challenged by anatomical variability of the tree (Fig. 1), presence of veins in the immediate vicinity of the arteries, and presence of pathology. Hence, to improve the extraction of the coronary artery trees, algorithms for automatic refinement of the extracted coronary trees were proposed. Zheng et al.⁹ applied a statistical model to extract and refine the coronary artery tree built from the main branches. Thereafter, the authors used region growth and subsequent tracking to append the centerlines of the side branches to the tree graph. Cao et al.¹⁷ extracted the coronary artery centerlines by applying thinning to a vesselness image. Thereafter, the authors applied a model-based strategy to remove false positives (FPs) and to bridge discontinuities by triggering a minimum-cost path search. He et al.¹⁰ used a CNN to generate initial coronary artery segmentations and subsequently applied thinning to obtain the centerline points. To identify FPs, the authors used a point cloud-based neural network. In addition, a minimum cost path search was applied to connect all disjointed segments to the tree.

Algorithms for coronary artery centerline extraction or refinement often utilized data that included manually annotated reference centerlines for a limited number of arteries in a scan. Therefore, previous works evaluated individual centerlines and not the entire tree.¹⁸

Unlike coronary tree extraction performed by analysis of CCTA scans, automatic anatomical labeling of the coronary artery segments is typically applied to previously extracted coronary trees. Early works labeled the coronary artery tree segments through registration with a labeled reference tree.^{19–21} However, registration methods are limited in their ability to label a wide range of anatomical geometries and to deal with errors that may be present in automatically extracted trees. To address variations in the tree geometry, multiple methods employed carefully tuned geometrical and topological rules designed based on expert knowledge.^{19–21} While this enabled labeling of automatically extracted coronary artery trees with missing side branches,

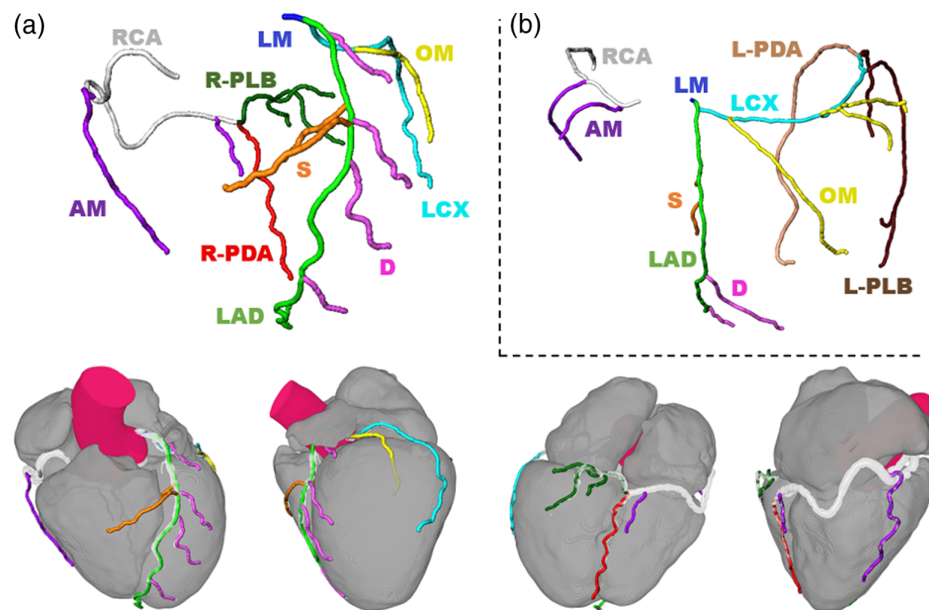


Fig. 1 (a) Coronary artery tree from the reference with right dominance type, as the posterior descending artery (PDA, red) is supplied by the RCA (white). Volume renderings in the bottom show the tree in its anatomical context from various angles. (b) Coronary artery tree with left dominance type, as the PDA (beige) originates from the LCX artery (light blue) (Video 1, MP4, 16.3 MB [URL: <https://doi.org/10.1117/1.JMI.11.3.034001.s1>]).

registration-based labeling of trees with potential discontinuities in the main branches remains an unsolved challenge. Alternatively, machine learning was used to label the coronary artery tree. This was achieved by analyzing manually segmented coronary artery lumen, i.e., by assigning anatomical labels to lumen voxels using CNNs²² or by labeling points sampled from the lumen using point cloud-based networks.^{22,23} Although analyzing the lumen segmentation enabled learning representations that include information about artery morphology, it has been shown that further processing is required to ensure good performance. Alternatively, machine learning was applied directly to the coronary artery centerline tree. For this, segments of the coronary artery tree were defined as centerlines between two bifurcations or between a bifurcation and an endpoint of the tree. These segments were characterized by features, which were then used as input to a model for assigning an anatomical label to each segment. Note that throughout this paper, “segment” refers to this definition as opposed to the anatomical segments as defined by Austen et al.,²⁴ which serve as the prediction targets in this work. Akinyemi et al.²⁵ labeled coronary artery tree segments by training a multivariate Gaussian classifier with subsequent post-processing in the form of geometrical and topological rules. Instead of using segments in isolation, Wu et al.¹⁵ used a tree-long short-term memory-based network (tree-LSTM) to model the coronary artery tree as an interleaved bag of sequences, achieving good performance without the need for further post-processing. However, the long-range representations learned by the tree-LSTM were likely sensitive to potential discontinuities in coronary arteries resulting from the tree extraction.¹⁶ Therefore, utilization of graph convolutional neural networks (GCNs), that apply convolutions directly on the structure of the coronary tree and enable aggregation of features from the neighboring segments, was investigated.^{16,26} In our preliminary work,²⁶ we demonstrated feasibility of using GCNs for anatomical labeling of coronary artery segments. Yang et al.¹⁶ showed that GCNs are able to improve robustness to synthetic gaps in the artery tree over the LSTM-based approach by Wu et al.¹⁵ Li et al.²⁷ compared the performance of different types of GCNs for coronary artery tree labeling. The aforementioned learning-based methods applied anatomical labeling to the manually extracted or manually corrected reference trees. Hence, these trees did not contain any FP segments or missing segments. Therefore, a method for anatomical labeling that maintains high performance despite potential inaccuracies from automatic tree extraction is desirable. To the best of our knowledge, this has not yet been investigated.

In this work, we present a novel approach for the fully automatic extraction and labeling of coronary artery tree segments in CCTA images. Our method first extracts the coronary artery tree with minimal false negatives (FNs) and represents it as an undirected tree graph. Subsequently, we utilize multi-resolution ensembles of GCNs for tree refinement. Last, we label the coronary artery tree using GCNs.

To the best of our knowledge, our study is the first work to combine automatic extraction and labeling of the coronary artery tree in CCTA without manual input while providing a comprehensive evaluation of the method's performance on complete coronary artery trees.

2 Materials and Method

We propose a method that automatically extracts the coronary artery tree by iteratively tracking automatically placed seed points. Subsequently, an ensemble of GCNs is used to refine the extracted tree and to label its segments (Fig. 2).

2.1 Data

This study includes two data sets: set 1 contains CCTA scans of 80 patients (age 47 to 79 years) acquired in the Onze-Lieve-Vrouweziekenhuis (Aalst, Belgium) with a Siemens Somatom Definition Flash CT scanner. Set 2 contains CCTA scans of 38 patients (age 48 to 85 years) acquired in the Amsterdam University Medical Center location AMC (Amsterdam, the

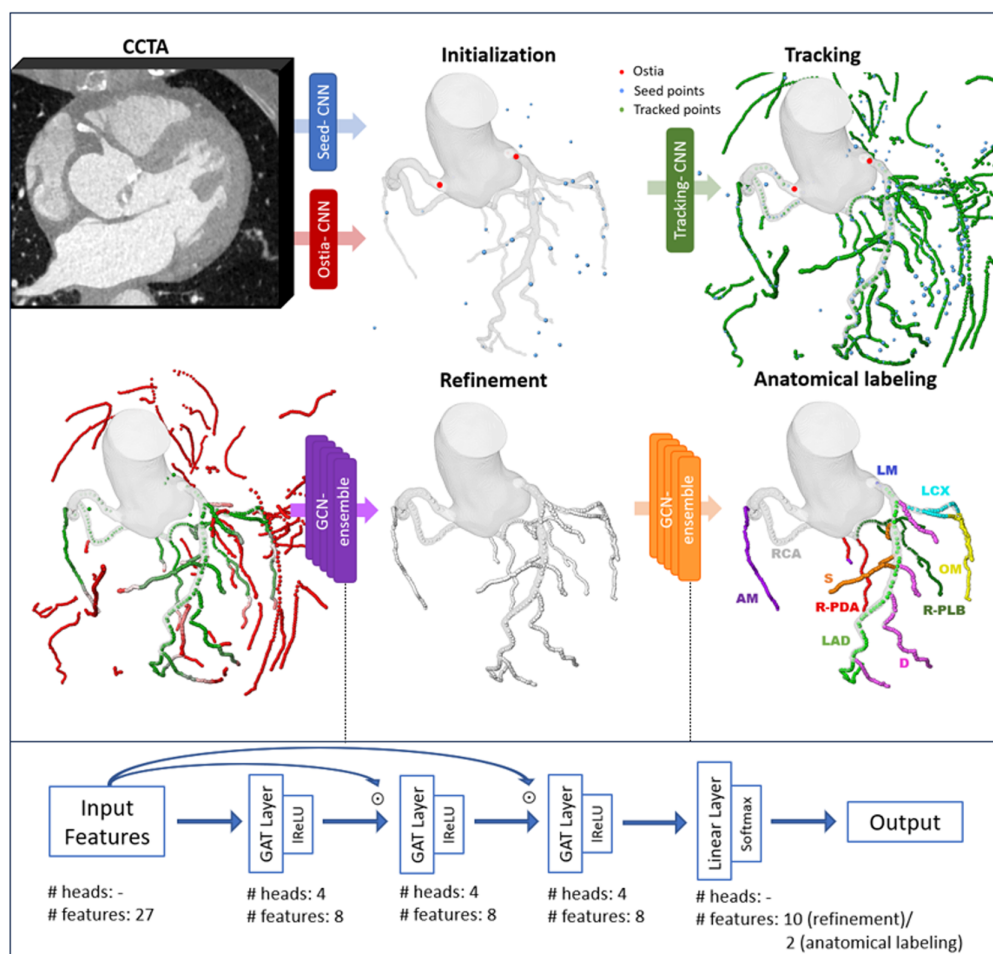


Fig. 2 Illustration of the proposed method. After initialization with the seeds and the ostia, the tree is tracked iteratively. Thereafter, an ensemble of GCNs is applied to refine the initially extracted tree (probability for removal of nodes indicated by red coloring). Finally, another ensemble of GCNs labels the anatomical segments. The architecture of the GCNs is shown at the bottom, where $\odot$ denotes concatenation.

Netherlands) with a Siemens Somatom Force CT scanner. The need for informed consent was waived by the respective local institutional review board. The contrast medium was injected using a flow rate of 4 to 6 mL/s for a total of 33 to 90 mL iopromide (Ultravist 300 mg I/mL, Bayer Healthcare, Berlin, Germany), depending on the patient weight and test bolus images.²⁸ Tube voltages between 70 and 120 kVp and tube currents between 71 and 901 mAs were used. All images were reconstructed to an in-plane resolution ranging from $0.22 \times 0.22 \text{ mm}^2$ to $0.83 \times 0.83 \text{ mm}^2$ with 0.5 to 0.9 mm slice thickness and 0.3 to 0.5 mm slice increment depending on the patient size. Nine and five images were excluded from sets 1 and 2, respectively, due to severe imaging artifacts. These artifacts included severe step-and-shoot artifacts (8), heart motion artifacts (3), artifacts caused by metal implants (2), and an incomplete field of view (1). After exclusion of scans with insufficient quality, in total 104 scans remained.

2.2 Reference Standard

To train and evaluate coronary artery tree extraction and labeling, completely identified reference trees are required. CT can only visualize sufficiently large coronary arteries, typically with a diameter greater than 1 mm, that are filled with contrast agent. Therefore, our reference includes all visible parts of the coronary artery trees, including poorly visible and irregularly appearing branches (e.g., septal branches or posterolateral branches). However, extraction of such detailed trees is very laborious. In addition, training an automatic method that is able to assign anatomical labels to such detailed trees requires a large amount of reference data. Therefore, to make the manual workload feasible, centerlines of the coronary artery tree were defined semi-automatically and thereafter manually corrected. The initial automatic artery centerline extraction was performed using our previously developed method¹² that tracks a centerline from an automatically or manually initialized seed point in the artery until its most distal point and its ostium. Ostia were automatically detected by a CNN developed as part of the same method and manually corrected when necessary. The automatic centerline extraction resulted in proximally overlapping centerlines. Hence, to create a reference tree from the separate coronary centerlines, we first randomly selected a centerline and subsequently added all remaining centerlines while removing overlapping parts automatically. The result was inspected and manually corrected when necessary.

All centerline points in the graph were manually assigned an anatomical label according to the model introduced by the American Heart Association.²⁴ This consisted of the following classes: left main (LM), left anterior descending (LAD), left circumflex (LCX), right coronary artery (RCA), diagonal (D), septal (S), obtuse marginal (OM), acute marginal (AM), right posterior descending artery (R-PDA), and right posterolateral branch (R-PLB). Given that the segments that predominantly occur in the rare left dominant systems [i.e., the left posterior descending artery (L-PDA) and the left posterolateral branch (L-PLB)] were scarce in our dataset, the L-PDA segments were labeled as LCX, and the L-PLB segments were labeled as OM. Furthermore, whereas application of machine learning to tree labeling requires segments to be defined and characterized prior to inference, subdivision of the three main arteries into proximal, medial, and distal segments can require landmarks that do not automatically follow from the tree topology. Therefore, in line with previous learning-based works,^{15,16,25,26} a single label was assigned to each of the three main arteries. Table 1 shows the average number of segments per class per coronary artery tree, stratified across two bins for in-plane and through-plane image resolution, respectively. Although most of our scans shared a similar resolution, the impact of image resolution remains evident, with higher-resolution images typically resulting in a greater number of segments across most classes.

2.3 Automatic Tree Extraction

We extract the coronary artery tree in two subsequent stages. First, we extract a tree using a CNN-based graph tracking method that initializes seed points and tracks the coronary arteries from these seed points. Thereafter, we refine the tree using an ensemble of GCNs, yielding the final coronary artery tree.

Our tracking method aims to represent the coronary tree as an undirected tree graph. Each point in the centerline corresponds to a node in the graph and the connections between centerline points are represented by undirected edges.

Table 1 Average number of segments per class per coronary artery tree in the dataset used to train the GCNs and to evaluate our method.

	In plane		Through plane	
	Low 50%	High 50%	Low 50%	High 50%
LM	0.98	1.06	0.98	1.05
LAD	4.90	5.64	4.98	5.41
D	2.33	2.47	2.23	2.51
S	4.62	3.77	4.18	4.28
LCX	3.05	3.45	3.30	3.18
OM	4.74	4.74	5.00	4.56
L-PDA	0.38	0.06	0.05	0.38
L-PLB	0.12	0.02	0.02	0.11
RCA	3.81	4.11	4.00	3.90
AM	4.76	4.64	4.52	4.84
R-PDA	1.55	1.51	1.66	1.44
R-PLB	3.17	3.64	3.32	3.43
Mean spacing	0.4	0.35	0.41	0.38

Please note that segments correspond to centerlines among bifurcations, leaf, and root points. The statistics are stratified across two resolution bins to evaluate the dependency on image resolution. For the thresholds between the bins (low 50% and high 50%), the medians of the resolutions were calculated, which equaled 0.39 mm for in-plane resolution and 0.40 mm for through-plane resolution. While the majority of our scans had a similar resolution, the effect of image resolution is still visible as for most classes higher-resolution images yield on average more segments.

2.3.1 Tree initialization

The tree graph is initialized by automatically placing seed points in the coronary arteries and left and right coronary ostia.¹² We propose to subsequently build the tree graph directly during coronary artery extraction by simultaneously tracking coronary centerlines from the identified seed points. In the process, we iteratively append new points in the coronary arteries to the tree graph. This way, we prevent redundantly tracking sections multiple times and reduce computational redundancy.

For initialization, i.e., for identifying seed points and the location of the coronary ostia, we use two fully CNNs (seed-CNN and ostia-CNN), described by Wolterink et al.¹² To make the networks invariant to differences in image resolution, CCTA scans were resampled to a uniform resolution of 1 mm. The seed-CNN and ostia-CNN predict for each voxel the negative exponential of the distance to the nearest coronary artery centerline or ostia, respectively. This renders a heatmap-like prediction map indicating where the coronary arteries and ostia are located. Thereafter, seed points are identified as local maxima from the predicted heatmaps. For further details, we refer to Wolterink et al.¹²

2.3.2 Tree tracking

To add new nodes to the tree, we employ a CNN (tracking-CNN) as described by Wolterink et al.¹² The tracking-CNN receives a three-dimensional (3D) image patch at the seed point location and predicts a direction in the form of binary outputs for discrete, evenly spaced locations on the unit sphere, as well as the radius of the coronary artery. To obtain a new point, a step is taken into the predicted direction with the step size corresponding to the vessel radius prediction.

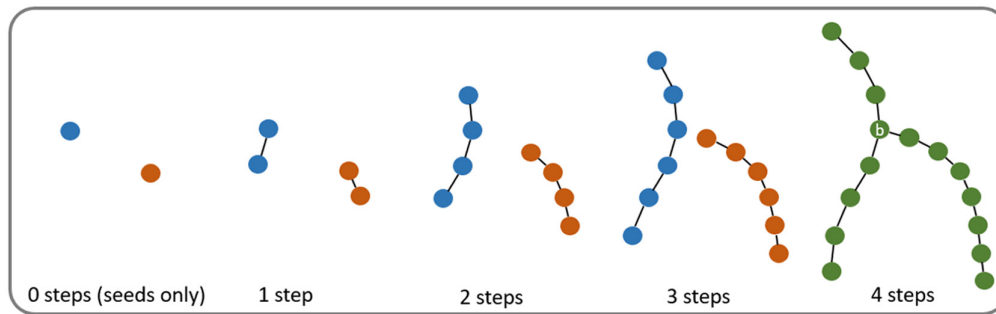


Fig. 3 Tracking of two seed points. As the two tracked sub-graphs (blue and red nodes) overlap after three steps, they are merged into a single connected graph (green nodes) in step 4. Note that this merging avoids redundant tracking of proximal nodes, which are located above the bifurcation (“b”).

The network also provides uncertainty estimates for direction predictions, calculated as the entropy over the output classes for different directions. Differences in resolution of CCTAs are handled by resampling the 3D image patch to a uniform resolution of 0.5 mm. For details on the method we refer to Wolterink et al.¹²

In the current work, we only track endpoints of the graph, i.e., nodes with fewer than two edges. Among these nodes, we only track those where the uncertainty of the direction prediction is below a certain threshold. Because nodes that are not connected to the ostia are less likely to reside in coronary arteries, we employ a different uncertainty threshold for nodes that are connected to the ostia than for all other nodes.

Simultaneous tracking of all seed points yields multiple sub-graphs. Hence, these tracked sub-graphs are merged when overlapping (Fig. 3). Therefore, a new node is created only when its location is not already occupied by another node, i.e., the node is not part of another sub-graph. The node overlap is assessed using the artery radius at each existing node as predicted by the tracker. Figure 4 illustrates building of the tree graph that includes the extraction of the seeds and subsequent artery tracking.

2.3.3 Tree refinement

During tree tracking, we aim to reach a high sensitivity to obtain a complete coronary artery tree. However, this may come at the cost of limited precision, i.e., FP extractions representing veins or other tubular structures. Therefore, to remove FPs from the extracted tree, we make use of an ensemble of GCNs. GCNs are a generalization of CNNs to arbitrary graph inputs. Therefore, GCNs enable learning combined representations of node features and connectivities directly from graphs. Hence, we need to define features that can subsequently be used by a GCN.

Feature extraction. To enable refinement of the initially extracted trees, we create artery segments by grouping adjacent centerline points of the tree graph. We characterize these segments using a set of features describing the location, orientation, geometry, and image appearance of the segments.

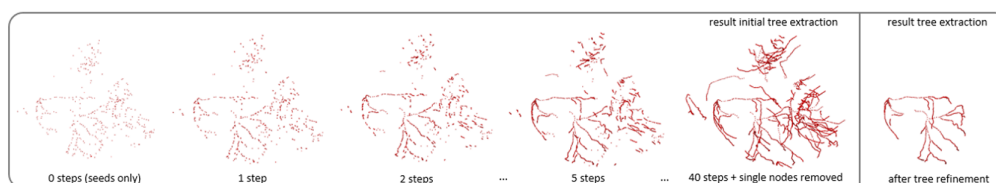


Fig. 4 Automatically extracted tree at multiple steps of the graph tracking. Note that the number of maximum steps is set to 40 or until no nodes remain to be tracked.

The location of the segment is described by the Cartesian coordinates of the segment's centerline points at the start, the end, and every quartile of the segment's length, with respect to the center of the left ventricle myocardium.²⁹ Two features are extracted that describe the segment's orientation. The first corresponds to the Cartesian coordinates of the normalized directional vectors between the segment endpoints. The second consists of the Cartesian coordinates of the normalized directional vectors between the first two centerline points in the segment. To describe the geometry of the vessel, the mean and standard deviation over the vessel radii of the centerline points in a segment are used, corresponding to the vessel size.

The appearance of the segment is characterized by its texture, derived using the outputs of the tracking-CNN and the seed-CNN (see Sec. 2.3). This is described by the mean and the standard deviation over the entropy values of the centerline points in a segment. The entropy value corresponds to the uncertainty of the tracker at that location, as described in Sec. 2.3. Intuitively, this indicates the extent to which the segment resembles a vessel. Similarly, the mean and standard deviation over the output values of the seed-CNN are calculated, extracted from the output map at the centerline point locations in a segment (Sec. 2.3). Given that the seed-CNN was trained on randomly extracted patches from all locations in the CCTA scans, these features may provide information that differentiates coronary arteries from other vessel-like structures.

Prior to the experiments, all features are normalized to zero mean and unit variance by determining the normalization parameters on the whole training set.

GCNs for coronary tree refinement. The coronary artery tree can be represented as a graph, where nodes are vessel segments that are connected by edges. As the graph can be directly used as input to GCNs, GCNs are particularly well suited to model relationships among these segments, as opposed to CNNs which operate on pixel level. Therefore, to distinguish real coronary artery segments (positive class) from other vessel-like structures (negative class), we perform binary classification using GCNs. GCNs enable learning combined representations of node features and connections using the initially extracted tree graph directly as input for classification.

In our preliminary work,²⁶ we showed that graph attention networks (GATs)³⁰ yield promising results for learning on coronary artery trees. In GATs, weights for the aggregation of features from neighboring nodes are learned end-to-end utilizing attention sub-networks. The aggregation function of a GAT is defined as $\vec{h}_{i'} = \sigma(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W} \vec{h}_j)$, with $\vec{h}_j$ as input features of nodes in the local neighborhoods $\mathcal{N}_i$ that are first transformed by the weight matrix $\mathbf{W}$. σ denotes a nonlinearity, for which we use the LeakyReLU function. To determine the weighting coefficients α_{ij} , attention subnetworks (heads) are used, which are parameterized by weight vectors $\vec{a}$, as

$$\alpha_{ij} = \frac{\exp(\sigma(\vec{a}^T [\mathbf{W} \vec{h}_i || \mathbf{W} \vec{h}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\sigma(\vec{a}^T [\mathbf{W} \vec{h}_i || \mathbf{W} \vec{h}_k]))}, \quad (1)$$

where $||$ denotes concatenation. As the attention mechanism enables the GAT to express the importance of neighboring segments for one another, they are likely a suitable choice for encoding local segment neighborhoods to refine our extracted coronary artery trees.

To leverage information about the geometrical structure of the coronary artery tree, a sufficiently large receptive field is needed. However, GCNs typically have a limited receptive field.³¹ Hence, we increase the effective receptive field using coarser graphs at lower resolutions as input. To create the coarse graphs, we group adjacent nodes of the initially extracted tree (fine graph) into segments.

Multi-resolution GCN ensemble. While grouping nodes increases the receptive field, it can also group true and false nodes into the same segment, e.g., when an artery centerline leaks into a vein, resulting in label ambiguity [Fig. 5(a)]. Therefore, we create multiple coarse graphs by grouping the centerline nodes into segments in two different manners. For the first coarse graph, we apply the conventional definition²⁶ in which segments are separated by bifurcations. To enable subdivision of these segments, we create additional graphs by dividing the

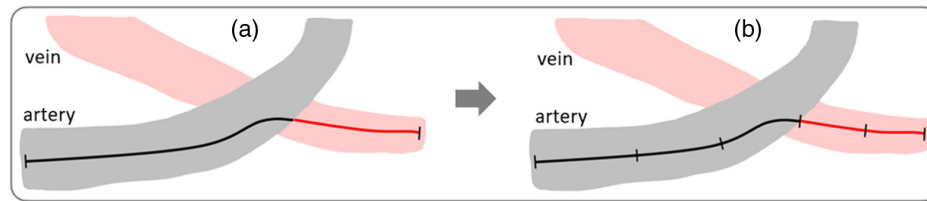


Fig. 5 (a) Leakage of the tracking of a coronary artery centerline (black) into a vein (red) leads to a segment including a centerline partially in a coronary artery and partially in a vein, rendering it ambiguous. (b) Dividing the segment into smaller sub-segments yields well-defined segments for tree refinement. This allows retaining the centerline in the artery and removing the centerline in the vein.

segments from the undivided graph above into smaller segments with higher resolutions. This subdivision is achieved by setting a maximum number of centerline points per segment, ensuring finer segmentation, and addressing the issue of label ambiguity caused by missing bifurcations or extractions. To enable information flow between all segments adjacent to a bifurcation, we establish the edges of the graphs by combining all possible pairs of involved segments. Furthermore, information flow from each node to itself is enabled by inserting an edge from each node to itself.

To combine the benefits of reduced label ambiguity (graphs with small segments) and the large receptive field (graphs with large segments), we employ a multi-resolution graph ensembling strategy. For this, we project the predictions for all graphs with different resolutions back to the fine graph. Ensembling was performed by taking the average over the output probabilities of all GCNs on the fine graph.

Whereas the GCNs in the ensemble are applied to graphs with different resolutions, the network architecture, shown schematically in Fig. 2, is identical for all GCNs. It is comprised of three GAT layers with dense connections from the input features to all consecutive GAT layers. In each GAT layer, four heads with eight encodings per head are used in combination with residual connections and dropout. In addition, the input features are concatenated to the input of each consecutive GAT layer as this improved performance in our preliminary work. Finally, binary output probabilities, corresponding to true segments of the coronary artery tree and other vessel-like structures, are predicted using a linear layer followed by the softmax activation function.

2.4 Coronary Artery Tree Labeling

We assign anatomical labels to the segments of the extracted coronary artery tree. Similar to tree extraction, we employ an ensemble of GCNs trained on graphs with different resolutions for anatomical labeling. We pose segment labeling as a multi-class classification task. By using GCNs, we enable the model to learn information about individual segments and the relation between these segments. Moreover, our multi-resolution ensemble approach mitigates the problems related to label ambiguity due to missing bifurcations, while ensuring that the receptive field of the method is large enough to model inter-segment relationships. The different resolution graphs are obtained by grouping centerline points into segments (Sec. 2.3.3).

To account for potential noise in the automatically extracted trees, instead of learning from reference trees, we use the graphs obtained from the output of our tree extraction stage directly as input for anatomical labeling. The network architecture for anatomical labeling is identical to the one for tree refinement (Sec. 2.3.3), with the exception that for anatomical labeling 10 output classes are used, one for each segment label (Sec. 2.2). Furthermore, the same input features are employed.

3 Evaluation

During artery extraction the step size, i.e. the distance between centerline points, varies with the diameter of the artery at that location. Therefore, to ensure that the evaluation is independent of the variation in the step size, the reference trees and the automatically extracted trees were resampled to a uniform distance between centerline points (0.25 mm, below the minimum radius) using trilinear interpolation.

Tree extraction was evaluated by defining true positive (TP), FP, and FN nodes as proposed by Schaap et al.¹⁸ For this evaluation, a TP node was defined as a node on the automatically extracted tree whose distance to the closest node on the reference tree was smaller than the radius of the artery at that node. An FP node was defined as a node on the automatically extracted tree that did not have a node on the reference tree within an artery radius distance. An FN node was defined as a node on the reference tree that did not have a node on the automatically extracted tree within an artery radius distance.

Anatomical labeling of the coronary artery tree was evaluated per class. Among nodes that were present in the reference tree and the automatically extracted tree, nodes with the correctly assigned anatomical label were counted as TP nodes. Nodes with differing labels in the reference trees and the automatically extracted trees were counted as FP nodes (incorrect automatic label assignment or node absent in the reference) and FN nodes (incorrect automatic label assignment or node not extracted).

Tree extraction and anatomical labeling were both evaluated in terms of recall

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (2)$$

precision

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

and *F1* score

$$\text{F1} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (4)$$

Metrics for all anatomical labeling classes combined were calculated by averaging over all classes (macroscopic averaging), to account for class imbalance. Furthermore, to evaluate parameter choices of the GCNs, the overall accuracy was calculated

$$\text{accuracy} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad (5)$$

including the segment classes and the background class, by weighting each node equally (microscopic averaging).

4 Experiments

To evaluate the performance of the proposed method and validate methodological choices, we performed a number of experiments.

We used CNN-based graph tracking that iteratively appends nodes to the coronary artery tree (Sec. 2.3). Common to all experiments, we integrated the CNNs for coronary artery tracking, and seed and ostia extraction by Wolterink et al.¹² into our method. For the initialization of the tracking, we used 400 seed points. We chose such a large number of seed points to ensure a high sensitivity. Moreover, we reduced the redundancy and ensured that seed points were not generated too close to one another by enforcing a minimum distance of 3 mm between seed points. For artery tracking, the input for the tracking-CNN was a $19 \times 19 \times 19$ image patch. We tracked nodes connected to the ostia if the entropy over the output directions, i.e., uncertainty, was below 0.9, and nodes disconnected from the ostia if the entropy was below 0.7. The tracker avoided backward tracking by masking directions within less than 60 deg of previously tracked directions.

Tracking terminated if no nodes remained to be tracked or if a maximum number of 40 steps—determined in preliminary experiments—was reached, preventing extensive FP extractions. In comparison to tracking proximal sections redundantly, the proposed graph tracking reduces the number of tracked nodes from an average 7667 to 4476 (42%). After tracking was finished, we removed all single nodes, as they presented seed points that had not been tracked. Tree tracking took on average 1 min per patient

To reduce label ambiguity (Fig. 5) and nevertheless learn from a large receptive field, we used a graph ensembling strategy based on coarse graphs with different resolutions. Specifically, in addition to the undivided graph, we also created graphs with maximally 5, 10, 20, and 30 centerline points per segment.

For training the GCNs and evaluating the final result of our method, we used the dataset described in Sec. 2.1, as it comprises fully annotated trees. To ensure unbiased testing, we randomly selected 25 scans from set 1 as a test set which were not used during development of our method. The remaining scans from set 1 (46) and set 2 (33) were used for training. We trained all GCNs for 500 epochs using the Adam optimizer. Unless specified differently, we set the learning rate to 0.001 for the first 300 epochs and to 0.0001 for the remaining 200 epochs. For dropout, we used a probability of 0.2. Following the default settings in Pytorch and Pytorch Geometric, all weights of the GAT layers were initialized randomly using Xavier initialization, and all biases were initialized with zeros.

Computations in this work were performed on a system with an Intel Xeon Gold 6128 CPU, featuring 12 cores at 3.4 GHz and 2 threads per core, and an NVIDIA GeForce RTX 2080 Ti GPU with 11 GB of RAM.

5 Results

We present the results for extraction of coronary artery trees, i.e., after tracking and refinement, and for coronary artery tree labeling.

5.1 Tree Extraction

To evaluate our tree extraction method, we performed experiments where we first extracted an initial tree and subsequently refined this tree (Sec. 2.3). We evaluated the performance using the 25 scans in the test set.

Tracking (Sec. 2.3.2) was performed for a total of 40 iterations in all but one case, where tracking stopped after 39 steps. As the tracker was trained on relatively small image patches around centerline voxels, it tracks coronary arteries, but it also tracks other vessel-like structures, such as coronary veins. Therefore, in this work, after tree tracking, we subsequently refined the tree by removing FPs, i.e., incorrectly identified segments, by applying automatic tree refinement (see Sec. 2.3.3).

Results for tree extraction are shown in Table 2 (columns 1 to 3). The performance was evaluated for GCNs based on coarse graphs with different segment lengths (i.e. different resolutions) and the ensemble. The network based on graphs with a segment length of 10 centerline points shows the best performance in terms of $F1$ score among individual GCNs. The best performance for tree extraction in terms of recall, precision, and $F1$ score was achieved by the ensemble of the GCNs. Therefore, the ensemble was used for tree extraction

Table 2 Results for our tree extraction (tree tracking and tree refinement) and our anatomical labeling with GCNs based on coarse graphs with different segment lengths and the GCN ensemble. As the ensemble enabled optimal performance for tree extraction, it was used to refine the graphs for evaluation of anatomical labeling. Best performances are indicated in bold.

Segment length	Tree extraction			Anatomical labeling		
	Recall	Precision	$F1$	Recall	Precision	$F1$
Points	0.82	0.86	0.83	0.72	0.72	0.72
Points	0.83	0.85	0.84	0.73	0.74	0.73
Points	0.83	0.84	0.83	0.72	0.75	0.73
Points	0.83	0.85	0.83	0.72	0.75	0.73
undivided	0.82	0.81	0.81	0.73	0.75	0.74
ensemble	0.84	0.87	0.85	0.73	0.76	0.74

in all further experiments unless stated differently. The analysis time for tree refinement was on average 30 s.

5.2 Anatomical Labeling

To enable the anatomical labeling method to handle to potential errors from tree extraction, instead of the reference trees, we used the automatically extracted trees as input to the GCNs for training. To train the ensemble for anatomical labeling, automatically extracted trees were represented by the same coarse graphs as for tree extraction, i.e., the undivided graph and graphs obtained using 5, 10, 20, and 30 as maximum centerline points per segment. Therefore, to set the reference for training the GCNs, we projected the anatomical labels from the reference trees onto the automatically extracted coarse graphs. During training of anatomical labeling, nodes that were present in the automatically extracted tree but not in the reference tree (FP nodes) were not backpropagated.

Table 2 (columns 4 to 6) lists the overall labeling performance of our method, i.e., anatomical labeling applied to the automatically refined trees. The results show that the performance of GCNs at different resolutions is very similar and that the ensemble slightly outperformed the individual networks. Therefore, the ensemble was used for all further experiments unless stated differently.

The performance of anatomical labeling per class is listed in Table 3. Please note that this table presents results classifying the 10 anatomical labels (i.e., assuming that the extracted tree contains only coronary artery segments). However, potential errors from tree extraction may impact the labeling performance and are therefore taken into account, as described in Sec. 3. Note that the reported performance varies significantly among the classes. The best performance, according to *F1* score, was achieved for the RCA branch, and the lowest performance was obtained for the small and diverse septal branches.

As illustrated in Fig. 6, our method successfully labeled coronary artery segments, even in the presence of disconnected tree segments caused by the automatic tree extraction. This suggests that the method can handle gaps, i.e., disconnected segments, in the tree. An additional illustration of our method on one subject from the test set can be found in Video 1. Tree labeling took on average 30 s.

Table 3 Results for the GCN ensemble used for anatomical labeling for different anatomical label classes as defined in Sec. 2.2. The resulting *F1* scores vary significantly between the classes.

Class	Anatomical labeling		
	Recall	Precision	<i>F1</i>
LM	0.71	0.70	0.70
LAD	0.88	0.85	0.86
D	0.73	0.73	0.73
S	0.50	0.57	0.54
LCX	0.68	0.81	0.74
OM	0.72	0.75	0.74
RCA	0.90	0.90	0.90
AM	0.83	0.86	0.84
R-PDA	0.66	0.72	0.69
R-PLB	0.68	0.70	0.69
mean	0.73	0.76	0.74

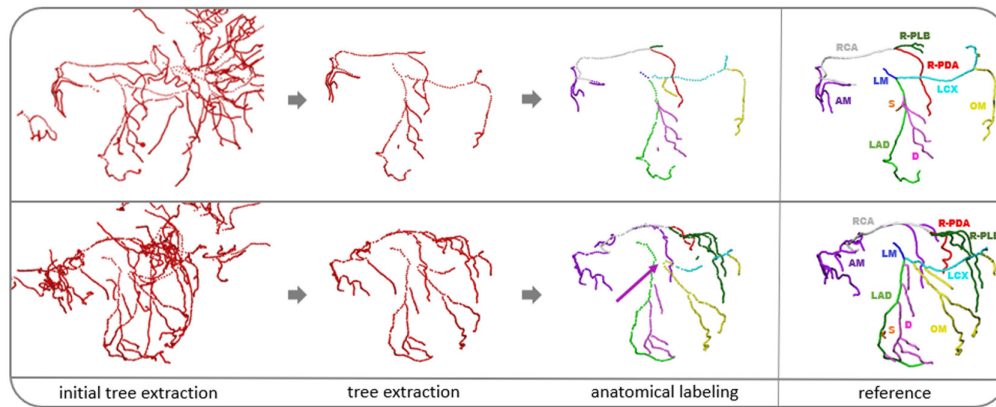


Fig. 6 Results of the initial artery tracking, tree refinement, and anatomical labeling for two subjects from the test set. The manual reference is shown for comparison. Whereas in the subject shown at the top, all correctly extracted segments were connected, in the subject shown at the bottom, excessive calcifications yielded a gap disconnecting the LCX from the LM (indicated by arrow). The ability of our method to extract the majority of centerlines in the disconnected sub-tree demonstrates the utility of our tree refinement step over excluding all centerlines disconnected to the ostia.

6 Ablation Study

We performed several experiments justifying our methodological and parameter choices. First, we evaluated the impact of the initialization settings of the graph tracking on the performance of our method. For a varying number of seed points and distances between seed points, we evaluated the performance of tree extraction before and after tree refinement. Results of the experiments on the test set are shown in Fig. 7. An increased number of seed points and a larger distance between them led to tracking of additional seed points less likely to be in coronary arteries. The results show that the performance saturates at 400 seed points with a minimum distance of 3 mm between them, as used in all our experiments.

Second, we investigate the influence of parameter choices in the network architecture and model optimization on the performance of the GCNs. For this purpose, we trained networks under various settings and evaluated extraction and labeling in terms of $F1$ score. Moreover, we evaluated the combination of extraction and labeling using the overall accuracy of the

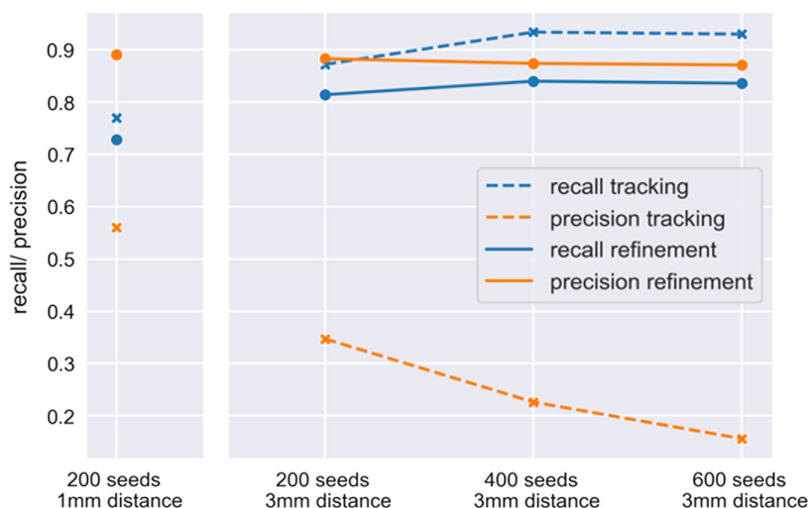


Fig. 7 Precision and recall for coronary artery tree tracking and refinement with varying numbers of initialized seed points. The first set of data points from the left corresponds to a minimum distance between seeds of 1 mm (as employed by Wolterink et al.¹²), whereas for the remaining experiments, 3 mm was used as a minimum distance.

Table 4 Performance of the GCN ensembles for coronary tree extraction and coronary segment labeling for different network architectures and parameters for model optimization. The parameter choices are compared in terms of $F1$ score ($F1$), separately for extraction and labeling, as described in Sec. 3. Furthermore, the accuracy was computed over the segment label classes and the background class combined using microscopic averaging, i.e., averaging over the nodes.

Tune	# layers	# filters	lr epoch < 300	lr epoch ≥ 300	$F1$ extraction	$F1$ labeling	Accuracy (segments + background)
Proposed	3	8	0.001	0.0001	0.86	0.74	0.94
Layers	1	—	—	—	0.84	0.72	0.93
	5	—	—	—	0.86	0.74	0.94
Filters	—	4	—	—	0.85	0.73	0.93
	—	16	—	—	0.86	0.74	0.94
Learning rates	—	—	0.01	0.001	0.86	0.74	0.93
	—	—	0.0001	0.00001	0.85	0.72	0.93

segment classes and background class combined. The results are listed in Table 4. The consistently high performance demonstrates the robustness of our method to the choice of these hyperparameters.

Third, we evaluated the influence of different types of features on the tree extraction and the anatomical labeling using GCN ensembles trained without each feature type (location, orientation, geometry, and image appearance of a segment). The results in Table 5 demonstrate that for both tree extraction and anatomical labeling, the proposed approach combining all feature types outperformed settings where individual types of features were omitted.

Fourth, to assess the value of information from local neighbors in tree refinement and anatomical labeling, we performed tree extraction and anatomical labeling using ensembles trained with only self-connections. Omitting information flow between neighboring nodes limited the receptive field to a single node. The proposed approach that used the neighborhood information outperformed the approach relying only on the segment at hand (Table 5).

Finally, to investigate whether identification of FP segments in the extracted coronary artery tree with the labeling network would improve segment labeling performance, we trained a labeling network that outputs 11 classes: 10 segment classes and one background class. This yielded

Table 5 Recall (recall), precision (prec), and $F1$ score ($F1$) of the second ablation experiment for tree extraction and anatomical labeling. Rows 1 to 4 list the results of ensembles trained by omitting each type of feature used in the proposed approach, i.e., characterizing location, orientation, geometry, and appearance of the segments. The fifth row lists the results of an ensemble trained on and applied to graphs containing only self-connections. The last row shows the results of the proposed approach. Best performances are indicated in bold.

	Tree extraction			Anatomical labeling		
	Recall	Prec	$F1$	Recall	Prec	$F1$
No location	0.81	0.84	0.82	0.60	0.61	0.59
No orientation	0.84	0.84	0.84	0.72	0.71	0.71
No geometry	0.84	0.86	0.85	0.71	0.72	0.72
No appearance	0.80	0.83	0.81	0.70	0.73	0.71
No neighborhood	0.81	0.86	0.83	0.70	0.74	0.72
Proposed	0.84	0.87	0.85	0.73	0.76	0.74

a recall, precision, and $F1$ score of 0.82, 0.87, and 0.84 for tree extraction and 0.72, 0.75, and 0.73 for tree labeling, respectively. The overall accuracy of the method was 0.93. Hence, using 11-class labeling of the coronary artery tree slightly underperformed compared with the proposed 10-class labeling approach.

7 Comparison with Previous Work

Several methods for the extraction of coronary artery centerlines or for anatomical labeling of the coronary artery tree were previously developed. Where possible, we compare experimentally, and otherwise, we report previous findings.

7.1 Tree Extraction

Given that reference annotation of complete coronary artery trees is extremely time-consuming, previous automatic methods were typically evaluated on a selected set of centerlines rather than the whole tree.¹⁸ However, this does not allow for evaluation of FP arteries. Hence, we only compare with a previous method that was evaluated on the whole tree (Table 6). Furthermore, we compare experimentally with our previous work¹² by applying it to our test data.

To the best of our knowledge, only the method by Salahuddin et al.,¹³ a CNN tracker, was evaluated on complete coronary artery trees (Table 6). Nevertheless, direct comparison is not possible because evaluation was performed on proprietary data and the dataset and software are not publicly available. The results show that in terms of recall, our method's performance approached that of Salahuddin et al.,¹³ and our method achieved a better $F1$ score. Note that compared with Salahuddin et al.,¹³ our training set was larger (79 versus 33 scans), and our trees contained on average twice as many branches (see Table 7 for characteristics of the trees used in our and previous work). This is likely because we extracted a larger number of short and thin branches.

Table 6 Comparison with the state of the art for tree extraction in terms of recall, precision, and $F1$ score. The performance of the method by Wolterink et al.¹² applied to our dataset and the performance as reported by Salahuddin et al.¹³ on their proprietary dataset are listed. The best average performance is indicated in bold.

	Recall	Precision	$F1$ score
Proposed	0.84	0.87	0.85
Wolterink et al. ¹²	0.68	0.85	0.75
Salahuddin et al. ¹³	0.87		0.80

Table 7 Characteristics of the trees used for reference for coronary artery tree extraction (row 1) and anatomical labeling of coronary artery trees. Rows 2 to 5 list statistics from previous work, while row 6 corresponds to the proposed work.

	Branches			Segments			Number
	Avg	Max	Min	Avg	Max	Min	Subjects
Salahuddin et al. ¹³	9	20	4				43
Wu et al. ¹⁵				22			436
Yang et al. ¹⁶	10	15	3	13	22	3	511
Li et al. ²²							526
Zhang et al. ²³							100
Proposed	18	32	11	33	61	20	104

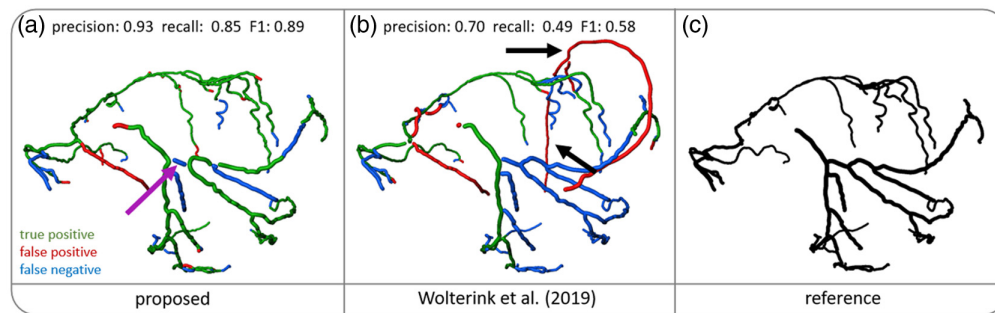


Fig. 8 Comparison of a tree from the test set extracted using the proposed method (a) with the same tree extracted using the method by Wolterink et al.¹² (b) and with the reference (c). Due to a tracking error in the proximal LCX [purple arrow (a)], the LCX and its side-branches were not extracted when using the method of Wolterink et al.¹² Furthermore, leakage of veins into arteries caused the veins to be part of the final result [black arrows (b)]. Our method enabled identifying a large proportion of the LCX and its side-branches while discarding most veins.

Furthermore, we applied our previous work¹² to the here used dataset. Given that previous work extracted separate centerlines and not the complete tree, we enabled comparison by merging all extracted centerlines into a tree prior to the evaluation.²⁶ The results show that the here presented method achieved higher recall, precision and $F1$ score (Table 6). This demonstrates that our current method extracts a substantially larger proportion of the coronary artery tree with less FP extractions. Figure 8 shows that our method is able to identify sub-trees that are separated from the ostia, i.e., due to a tracking error, as well as to discard veins that leak into arteries.

7.2 Anatomical Labeling

Several machine learning methods were previously developed for anatomical labeling of the coronary artery tree from CCTA. Previous works for anatomical labeling were evaluated on manually extracted or manually refined coronary artery trees. Using corrected trees as input likely results in higher performance, as no inaccuracies from tree extraction are present in training and testing. Therefore, for fair comparison, we additionally investigate the performance of our anatomical labeling in the same setting. Hence, we perform experiments using manually extracted trees (reference trees) as input (Table 8). As ambiguities in the reference trees are strongly reduced, GCNs based on low-resolution graphs likely outperform GCNs based on high-resolution graphs due to their enlarged receptive field. Therefore, to enable a fair comparison of our method on the reference trees, we separately list the performance of the GCN based on undivided segments (lowest resolution). In Table 9, the performance attained by the most successful previous deep learning-based methods is listed, demonstrating that the performance of our method is on par with the state of the art. However, note that also here, direct comparison is not possible as the methods we compare with used different datasets of substantially different sizes and characteristics. Table 7 lists these characteristics for previous graph-based methods.^{15,16} Methods operating on the lumen segmentation did not report graph characteristics.^{22,23} A smaller number of subjects and a larger number of segments per tree likely increased the difficulty of anatomical labeling in our work in comparison with previous graph-based methods.

8 Discussion

We presented a method that automatically extracts the coronary artery tree and assigns anatomical labels to its segments in CCTA scans. The results demonstrate that our method enables extraction and anatomical labeling of the complete coronary artery tree, including small side branches, at the expense of a low number of FPs.

For extraction of the coronary arteries, previous works tracked each centerline individually^{12,32} and merged them afterward.^{16,26} Instead, we employed graph tracking that reduces computational redundancy by tracking overlapping parts only once. Furthermore, previous works tracked the centerlines from distally located seed points until the ostium. Instead, our method tracks all seed points simultaneously. This way, we do not discard centerlines that do not

Table 8 Performance of our method for different evaluation settings in terms of $F1$ score for different anatomical label classes (Sec. 2.2) and the mean over all classes. In column 2, the performance of the proposed method, i.e., fully automatic coronary tree extraction and labeling, is shown. Columns 3 to 4 list the results obtained when training and evaluating our segment labeling on reference trees (thus not the automatically extracted trees). Note that in column 4, instead of the ensemble, only the GCN based on the undivided segments (without a maximum segment length) was used.

Setting network	Automatically extracted trees ensemble	Reference trees	
		Ensemble	Only undivided segments
LM	0.70	0.85	1.00
LAD	0.86	0.95	1.00
D	0.73	0.94	0.98
S	0.54	0.96	0.99
LCX	0.74	0.85	0.87
OM	0.74	0.90	0.93
RCA	0.90	0.95	0.97
AM	0.84	0.98	0.99
R-PDA	0.69	0.84	0.88
R-PLB	0.69	0.87	0.92
mean	0.74	0.91	0.95

Table 9 Comparison with state-of-the-art for anatomical labeling in terms of $F1$ score for different anatomical label classes (Sec. 2.2) and the mean over all classes. In column 2, the performance of the proposed method is given. To provide a fair comparison with previous works, these results were obtained by training and evaluating our method on reference trees (thus not the automatically extracted trees). Furthermore, in line with previous works, instead of the ensemble, only the GCN based on the undivided segments (without a maximum segment length) was used. Columns 3 to 6 list results of previous deep learning-based methods for anatomical labeling, applied to the coronary artery lumen segmentation^{22,23} or to the coronary artery centerline tree^{15,16} (as our proposed work). Note that training and evaluating our method on the reference trees, as opposed to the automatically extracted trees, enable performance on par with the state of the art. Best performances are indicated in bold.

	Proposed labeling based on undivided segments applied to manually corrected tree	Li et al. ²²	Zhang et al. ^{23a}	Wu et al. ¹⁵	Yang et al. ¹⁶	Li et al. ²⁷
LM	1.00	0.96	0.96	0.99	0.99	0.90
LAD	1.00	0.91	0.98	0.97	0.99	0.85
D	0.98	0.90	0.93	0.94	0.91	0.80
S	0.99		0.91	0.82	0.96	0.71
LCX	0.87	0.90	0.99	0.92	0.98	0.89
OM	0.93	0.86	0.92	0.88	0.91	0.80
RCA	0.97	0.95	1.00	0.95	0.99	0.90
AM	0.99		1.00		0.99	0.80
R-PDA	0.88	0.85	0.97	0.78	0.94	0.51
R-PLB	0.92	0.94	0.99	0.87	0.95	0.85
mean	0.95	0.91	0.97	0.87	0.96	0.80

^aEvaluation on points sampled evenly within the artery lumen, hence giving greater weight to thicker arteries.

lead to the ostium, e.g., due to an erroneous choice of tracking direction at a bifurcation. We observed that increasing the number of seed points for tree extraction leads to saturation of recall at a value slightly below 1 (Fig. 7). Even though this value is high, the lack of further improvement in recall might have been caused by missed initialization of seed points in very small segments or tracking failures, e.g., due to pathology or nearby veins. Tracking through sections affected by pathology or imaging artifacts where the tracking is challenging might be further improved by initializing multiple trackers at slightly different initial locations such as recently proposed by van Harten et al.³³

Our approach tracks each node only a maximum of two times. This means that at points connected by more than two branches, only two branches can be tracked, and the remaining ones require their own seed point (Fig. 3). Previous work demonstrated that detecting bifurcations enabled tracking of all outgoing branches from bifurcations.¹³ This allowed tracking the entire coronary artery tree from a single seed point in the left and right branches of the coronary artery tree. However, relying on a single seed point per coronary branch may cause a substantial number of FNs caused by local pathology or imaging artifacts. Future work could investigate whether the recall of our method can be further improved by combining tracking of a high number of seed points with bifurcation prediction and subsequent tracking of all involved branches.

In our tree extraction, we first tracked an overcomplete tree and thereafter refined the tree. By designing the tree tracking specifically to have a high recall, a minimal number of arteries was missed, however, at the expense of a low precision. The results show that, despite the high FP rate in tracking, the tree refinement was able to extract the coronary artery tree with high precision.

The tree refinement is based on graphs constructed at high resolution (small segments) and low resolution (large segments). To combine the respective benefits, i.e., reduced label ambiguities (Fig. 5) and a large receptive field, a multi-resolution ensemble approach is proposed. Separate evaluation of the GCNs in the ensemble showed that the performance was higher for GCNs based on high-resolution graphs than for GCNs based on low-resolution graphs. This implies that for tree extraction, avoiding ambiguities is more important than the use of a large receptive field. The observed good performance despite a small receptive field underlines the descriptive power of local information. This is in line with our ablation experiment, which showed that minimizing the receptive field to a single node (Table 5) allows for retaining a high-performance level. However, in some cases, location and other local features describing orientation, geometry, and appearance of veins may be very similar to arteries. In such cases, global geometry and global appearance may enable discriminating arteries from veins. Global characteristics can be exploited by GCNs based on low-resolution graphs, as the larger receptive field enables gathering information from distant segments. Improved performance of our ensemble over the individual GCNs shows that global information has added value for tree extraction.

Prior to labeling coronary segments, we assume that only valid coronary centerline segments are extracted. Nevertheless, the extracted tree may contain FP segments. To evaluate the potential benefit of adding the background class to tree labeling classification, we posed the tree labeling as an 11-class classification problem (10 segment classes and one background class). The results demonstrated that 11-class classification led to slightly lower performance than the 10-class classification. Retrospective analysis showed that 11-class classification erroneously labeled a number of true coronary centerline nodes as background.

As with tree refinement, for anatomical tree labeling, we employed an ensemble of GCNs based on high- and low-resolution graphs. Our experiments show that the ensemble of GCNs enables accurate anatomical labeling of the vast majority of the automatically extracted coronary artery tree segments. Opposite to what we found in tree refinement, in anatomical labeling, GCNs based on low-resolution graphs (large segments) outperformed GCNs based on high-resolution graphs (small segments, Table 2). This implies that global information obtained with a large receptive field is essential for anatomical labeling and potential label ambiguities in segments may play a less important role. Nevertheless, application to the automatically extracted trees led to the ensemble slightly outperforming all separate GCNs (Table 2). This demonstrates that anatomical labeling of the automatically extracted coronary artery tree also benefits from higher-resolution graphs.

When applying our anatomical labeling to the reference trees (manually extracted by experts) instead of the automatically extracted trees, the performance increased substantially.

As segments of the reference trees are guaranteed to be complete, in this setting, the GCN with the largest receptive field outperformed the ensemble of GCNs. The achieved results (0.95 *F1* score) demonstrate that the performance of our labeling is on par with the state-of-the-art for anatomical labeling (0.87 to 0.97 *F1* score,^{15,16,22,23}).

The performance of anatomical labeling varied over the different classes. On the one hand, this is likely influenced by errors in tree extraction being more pronounced for certain classes than others. For instance, septal branches (class S) were frequently missed, which impacted the respective labeling performance. These branches are typically short and thin and hence also challenging to identify visually. However, for the same reason, vessels of class S are invasively treated only in exceptional cases and usually play a subordinate role in the perfusion of the heart.³⁴ Hence, these errors are of limited clinical relevance. Moreover, the retrospective analysis revealed that the moderate performance for the LM segments may have been associated with tracking beyond the ostium into the aorta, causing the ensemble for tree refinement to discard the entire branch in some cases. When using the reference trees as input, the performance is on par with the state of the art, yielding substantial improvements for these segments. This indicates that improving tree extraction would improve performance of anatomical labeling for these segments. On the other hand, some segments can be challenging to distinguish as their typical locations are similar. This is, for instance, the case for the R-PDA and R-PLB and is likely the cause for the lower-than-average performance in these segments. To improve performance in these segments, additional relative features or rule-based post-processing could be applied.

The results in this work indicate that the ability to correctly label segments in the presence of gaps in the tree comes at the cost of reduced performance. Occasionally occurring disconnected segments might have hampered information flow in the GCNs. Future work might address these issues by improving the CNN tracker. For example, Salahuddin et al.¹³ reported performance gains for CNN trackers taking multi-resolution input. Moreover, a second tracker could be trained specifically for difficult sections, affected by severe pathology. In addition, our proposed system is not end-to-end. Combining tracking, refinement, and labeling into a single network would enable the flow of gradients from refinement and labeling into tree tracking, potentially improving the performance of the entire system. However, even in the case of such improvements, small gaps in segments affected by severe pathology will still likely occur. Therefore, dealing with disconnected parts of the coronary artery tree, as proposed in this work, may remain advantageous. Using GCNs for tree refinement and anatomical labeling, we enable analysis of branches that are not connected to the ostia, instead of discarding them, such as in previous works.^{12,32} Future work may investigate whether the ability of the proposed approach to deal with disconnected parts may be improved.

9 Conclusion

We presented a deep learning approach for fully automatic extraction and labeling of the coronary artery tree. The results demonstrate that our tree extraction improves upon the state of the art and that our tree labeling maintains high performance despite inaccuracies in the tree graphs caused by automatic extraction. To the best of our knowledge, this work presents the first combination of automatic extraction and labeling of the coronary artery tree in CCTA without manual input that is evaluated using complete coronary artery trees as reference. The results indicate that the proposed approach achieves high *F1* scores, paving the way to fully automatic reporting of CAD.

Disclosures

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Code and Data Availability

The data that support the findings of this article are not publicly available due to privacy.

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Biographies of the authors are not available.